



Time: 5 minutes

Outcome: Calculate tax wedges, buyer and seller prices, quantity effects, and revenue.

Difficulty: Intermediate

Tax Wedges and Revenue

THE CORE IDEA

A per-unit tax creates a wedge between the price buyers pay and the price sellers receive. The wedge reduces the quantity traded because some exchanges whose benefits formerly exceeded their costs no longer occur. Tax revenue equals the per-unit wedge multiplied by the post-tax quantity, while the lost mutually beneficial trades create deadweight loss.

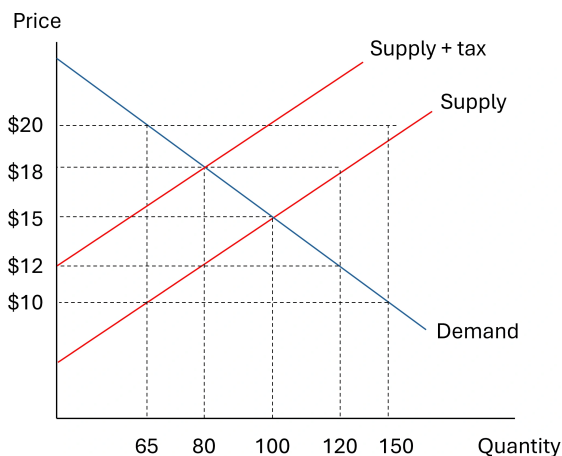
HOW TO RECOGNIZE IT

- Identify the tax wedge between the price buyers pay and the price sellers receive.
- Use the post-tax quantity, not the original equilibrium quantity, when measuring revenue.
- Tax revenue equals the per-unit tax multiplied by the quantity traded after the tax.
- A tax normally reduces quantity traded and creates deadweight loss from forgone exchanges.

WATCH OUT

Tax revenue equals the per-unit tax times the quantity actually traded after the tax. Use the \$6 wedge and 80 units here; do not multiply by the original untaxed quantity.

WORKED EXAMPLE: CALCULATING TAX REVENUE



The untaxed equilibrium is \$15 at 100 units. After the tax, buyers pay \$18, sellers receive \$12, and quantity falls to 80. The tax wedge is $\$18 - \$12 = \$6$ per unit, so government revenue is $\$6 \times 80 = \480 .

CHECK YOURSELF

Buyers pay \$18 after a tax and sellers receive \$12. What is the tax per unit?